Nathaniel Felsky-Deelstra

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Education

Queen's University, BASc Electrical and Computer Engineering — Sept 2025 – Expected 2029

Engineering Teams

Simulation Lead, aQuatonomous — May 2026 – Present

- Built a Gazebo/ROS 2 Jazzy simulation of the autonomous surface vessel with 4 sensors (LiDAR, cameras, GPS, IMU), letting the software team test navigation without in-water access.
- Lead 13 students running weekly standups, reviewing code, and assigning tasks.

Electrical Team Member, aQuatonomous — September 2025 – May 2026

- Reduced telemetry interference by modelling magnetic fields from the power supply and repositioning the radio, sensors, and wiring.

Projects

Modern Slavery Hotline

- Built a voice hotline that educates at-risk workers about forced labour and their rights, using Twilio, Google Speech, Claude, and Resend.
- Cut response time from 10 s to under 3 s without losing answer quality.

Water Pre-Treatment Plant

- Designed a turbidity-reduction system, achieving more than 2500 NTU improvement in under 8 minutes using coagulation and press filtration.
- Designed CAD parts and gear ratios for the mechanism, and automated pump and sensor control with Arduino.

Smart Mirror

- Built a Raspberry Pi 3A+ smart mirror (two-way acrylic) showing weather, calendar, and now-playing data via APIs in Python/JS.

Jacket Planner

- Built a three.js web app for Queen's students to design jackets and preview them in 3D, used by 100+ students.

Experience

Social & Economic Policy Researcher, Canadian Labour Congress — June 2026 – July 2026

- Researched the societal impacts and growth outlook of modular construction technology and AI.
- Interviewed professors and compiled data to produce an annotated bibliography used to advise the congress.

Leadership

Undergrad Curriculum Committee Rep, Department Delegate, Queen's ECE Department — September 2026 – Present

- Attend biweekly meetings with professors and delegates to discuss curriculum and department changes on behalf of ECE undergraduates.

Elected Year Representative, Queen's ECE Executive Club — September 2025 – April 2026

- Represented first-year Computer Engineering students on the ECE exec, collecting feedback and raising issues with the team.
- Designed the first-year program patch and helped run community events, including ECE Night and intramural activities.

Skills

Languages: C, C++, C#, Python, JavaScript
Software: Fusion 360, Git, Gazebo, ROS 2 Jazzy, KiCad
Hardware: Soldering, Circuit analysis, Arduino, Raspberry Pi